

Temporal Difference Learning

Temporal Difference Learning

- Bootstrap using prior knowledge like DP

$$\begin{aligned}
 v_{\pi_k}(s) &= E[R_{t+1} + \gamma v_{\pi_k}(S_{t+1}) | S_t = s] \\
 &= \sum_a \pi(a|s) \sum_{s', r} (r + \gamma v_{\pi_k}(s')) p(s', r | s, a)
 \end{aligned}$$

Bootstrap as in use our knowledge of the values of next states, for time/skye $t+1, t+2, \dots$

- However, use sample estimates of expectation like MC
 - Learn from experience
- Method, unlike MC, is truly online

RV X .

Can't calculate $E[X]$ exactly.

We used sample mean of X to approximate $E[X]$.

We drew $n \geq 1$ samples $X_1, X_2, \dots, X_n$, where X_i has the distribution of X .

$\Rightarrow X_i$ is an estimate of $E[X]$

We want to estimate

$$\underbrace{E[R_{t+1} | S_t = s]}_{\text{Reward for the time-step}} + \gamma \underbrace{E[V_{\pi}(S_{t+1}) | S_t = s]}_{\text{Bootstrapping}}$$

$$= E[R_{t+1} + \gamma V_{\pi}(S_{t+1}) | S_t = s]$$

The rv $R_{t+1} + \gamma V_{\pi}(S_{t+1}) | S_t = s$ is a sample mean of the expected value

$$E[R_{t+1} + \gamma V_{\pi}(S_{t+1}) | S_t = s],$$

wherein we used exactly one sample to estimate the expected value

Update step:

$$\begin{aligned} \underline{V_{\pi}(s)} &= (1-\alpha) \underbrace{V_{\pi}(s)}_{\uparrow} + \alpha \left(\underbrace{r_t + \gamma V_{\pi}(s')}_{\downarrow} \right) \\ &= V_{\pi}(s) + \alpha \underbrace{(r_t + \gamma V_{\pi}(s') - V_{\pi}(s))}_{\text{TD-error}} \end{aligned}$$

Averaging in MC

You ran an episode, obtained a return and used it to update the value function (expected return starting in the state S_t)

4: Generate an episode using π . We get

5: $S_0, A_0, R_1, S_1, A_1, R_2, \dots, S_T, R_T$

6: $G = 0$

7: **for** $t = T - 1, T - 2, \dots, 0$ **do**

8: $G = G + \gamma R_{t+1}$

9: $v_{\pi}(S_t) = (1 - \alpha)v_{\pi}(S_t) + \alpha G$

sample return value

Temporal Difference.... You are Impatient

- Perform action at time t ,
- To update value of state at t
 - Use the resulting reward,
 - and Bootstrap using current value of the next state!

$$\begin{aligned}
 v_{\pi}(S_t) &= (1 - \alpha)v_{\pi}(S_t) + \alpha(R_{t+1} + \gamma v_{\pi}(S_{t+1})) \\
 &= v_{\pi}(S_t) + \alpha(R_{t+1} + \gamma v_{\pi}(S_{t+1}) - v_{\pi}(S_t))
 \end{aligned}$$

Temporal Difference

$$v_{\pi}(S_t) = v_{\pi}(S_t) + \alpha(R_{t+1} + \gamma v_{\pi}(S_{t+1}) - v_{\pi}(S_t))$$

- We want to move the current value of the state closer to the *target*
- Do so by shifting the current value by the *TD-Error*

Temporal Difference Error and MC Error

- TD Error

$$\delta_t = R_{t+1} + \gamma v_{\pi}(S_{t+1}) - v_{\pi}(S_t)$$

- MC Error

TD target *current value / estimate*

$$\delta^{\text{MC}} = R_{t+1} + \gamma R_{t+2} + \dots + \gamma^{T-1} R_T - v_{\pi}(S_t)$$

MC target

TD vs. MC (Journey From Work to Home)

- Your chosen policy takes you from work (State 0) to home (3) via Kalkaji (1) and Hauz Khas (2)
 - Reward at a state is the time taken to reach the next state
- You start from Work (State 0), your current estimate to home is 30 minutes

$$v_{\pi}(0) = 30$$

TD vs. MC (Journey From Work to Home)

- Your chosen policy takes you from work (State 0) to home (3) via Kalkaji (1) and Hauz Khas (2)
- You start from Work (State 0), your current estimate to home is 30 minutes

$$v_{\pi}(0) = 30 \quad \checkmark$$

- You reach Kalkaji 15 minutes later

$$\hookrightarrow R_{t+1} = 15$$

TD vs. MC (Journey From Work to Home)

$$\alpha=1, \gamma=1$$

- You start from Work (State 0), your current estimate to home is 30 minutes

$$v_{\pi}(0) = 30$$

- You reach Kalkaji 15 minutes later

$$R_{t+1} = 15$$

- Suppose

$$v_{\pi}(1) = \underline{25}$$

MC (MC):

$v_{\pi}(0) = 30$. Not updated.

TD: $v_{\pi}(0) = v_{\pi}(0) + \alpha [15 + (1)(25) - v_{\pi}(0)]$

Using TD

- TD Target for state 1:

$$R_{t+1} + \gamma v_{\pi}(1) = 15 + 25 = 40$$

- Updated value of 0,
 - which is time total time you now expect to take work to home

$$\underbrace{v_{\pi}(0)}_{25} + \alpha(\underbrace{R_{t+1} + \gamma v_{\pi}(1)}_{40} - \underbrace{v_{\pi}(0)}_{25}) = 40$$

- TD Error

$$\delta_0 = 10$$

Using TD

- From Kalkaji to Hauz Khas you take an additional 15 minutes

$$\overbrace{R_{t+2}} = 15$$

- Suppose

$$\underline{v_{\pi}(2) = 7}$$

MC: stays at $v_{\pi}(0) = 30$.

- I can revise my target for state 0. What is it?

TD: $\left[\begin{array}{l} \text{New target} \\ \text{estimate of } v_{\pi}(0) \end{array} \right] \rightarrow R_{t+1} + R_{t+2} + v_{\pi}(2)$
 $= 15 + 15 + 7 = 37.$

Current estimate
 $v_{\pi}(0) = 40.$

 Revised is
 $40 + \alpha(37 - 40) = 37,$
 given $\alpha = 1$

Using TD

- From Kalkaji to Hauz Khas you take an additional 15 minutes

- Suppose

$$v_{\pi}(2) = 7$$

- I can revise my target for state 0. It is 37 and the TD error is $37 - 40 = -3$

Using TD

- Suppose Hauz Khas (state 3) to home takes 3 minutes

$$R_{t+3} = 3$$

- Suppose

$$\underline{v_{\pi}(3) = 1}$$

- What is my revised target for state 0?

$$\begin{aligned} & R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \gamma^3 v_{\pi}(3) \\ &= 15 + 15 + 3 + 1 \quad [\gamma=1] \\ &= 34. \end{aligned}$$

- What is the TD error

$$\underline{34 - 37 = -3.}$$

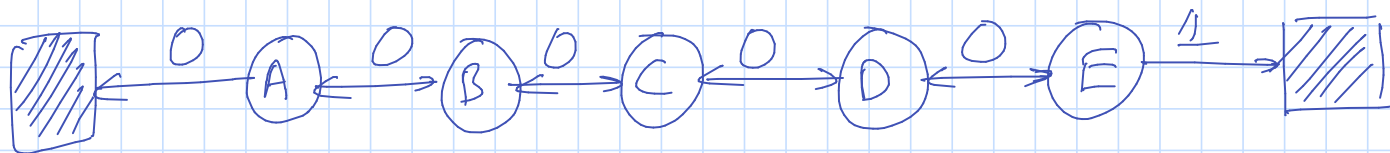
Using MC

- Target is known only once you reach home...
- MC error tells you
 - how bad your predictions were while your were driving from work to home

Graph the Errors

TD Prediction

- 1: **while** true **do**
 - 2: Initialize step $t = 0$ at beginning of a new episode
 - 3: Initialize $v_\pi(s), \forall s \in S$
 - 4: Randomly pick an initial state S_0
 - 5: Pick action A_0 as per $\pi(a|S_0)$
 - 6: **for** Step $t + 1$ in an episode **do**
 - 7: $v_\pi(S_t) = v_\pi(S_t) + \alpha(R_{t+1} + \gamma v_\pi(S_{t+1}) - v_\pi(S_t))$
 - 8: Pick action A_{t+1} as per $\pi(a|S_{t+1})$
-



Assume equi-probable left/right actions.

$$V_{\pi}(A) = \frac{1}{2}(0) + \frac{1}{2}(0 + V_{\pi}(B))$$

$$= \frac{V_{\pi}(B)}{2}$$

$$V_{\pi}(B) = \frac{1}{2}V_{\pi}(A) + \frac{1}{2}V_{\pi}(C)$$

$$V_{\pi}(C) = \frac{1}{2}V_{\pi}(B) + \frac{1}{2}V_{\pi}(D)$$

$$\rightarrow V_{\pi}(E) = \frac{1}{2} + \frac{1}{2}V_{\pi}(D) \quad \longleftrightarrow \quad V_{\pi}(D) = \frac{1}{2}V_{\pi}(C) + \frac{1}{2}V_{\pi}(E)$$

$$V_{\pi}(B) = 2V_{\pi}(A)$$

$$V_{\pi}(C) = 3V_{\pi}(A)$$

$$V_{\pi}(D) = 4V_{\pi}(A)$$

$$V_{\pi}(E) = 5V_{\pi}(A)$$

$$\Rightarrow 5V_{\pi}(A) = \frac{1}{2} + \frac{1}{2} 4V_{\pi}(A)$$

$$3V_{\pi}(A) = \frac{1}{2} \Rightarrow V_{\pi}\left(\frac{1}{6}\right)$$

OPTIMALITY OF TD(0).

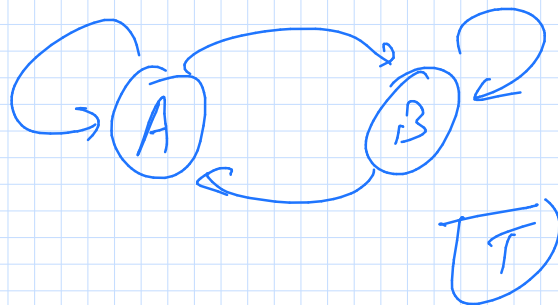
→ TD(0) results in "certainly-equivalent estimates" of the value function, given the data.

↳ If model captured by the data is accurate, TD(0) estimates converge to the true value function.

→ MC based estimate minimizes the squared error, given the data.

1st: A, 0, B, 0
2nd: B, 1
3rd: B, 1
4th: B, 1
5th: B, 1

} 5 episodes of data.



MC updates? $V_{\pi}(B) = \frac{4}{5}$, $V_{\pi}(A) = 0$

TD updates?

Model from data?

→ First Ep: $V_{\pi}(A) = 0$, $V_{\pi}(B) = 0$.

2nd: $V_{\pi}(A) = 0$, $V_{\pi}(B) = \frac{1}{2}$

3rd: $V_{\pi}(A) = 0$, $V_{\pi}(B) = \frac{2}{3}$

4th: $V_{\pi}(A) = 0$, $V_{\pi}(B) = \frac{3}{4}$

5th: $V_{\pi}(A) = 0$, $V_{\pi}(B) = \frac{4}{5}$.

Now update $V_{\pi}(A)$ using

bootstrapping: $V_{\pi}(A) = 0 + \frac{4}{5} = \frac{4}{5}$.